

Python Fundamentals — Variables, Keywords & Data Types

A practical reference guide covering Python variables, reserved keywords, and built-in data types.

1. Variables

A variable is a name that refers to a value stored in memory. Python is dynamically typed, meaning you don't declare a type — it is inferred automatically from the value assigned, and a variable can be reassigned to a different type later.

Creating variables

```
name = "Alice"      # string
age = 30            # integer
height = 5.6        # float
is_active = True    # boolean

# Multiple assignment
x, y, z = 1, 2, 3

# Same value to multiple variables
a = b = c = 0
```

Naming rules

- Must start with a letter or underscore (`_`), not a digit.
- Can contain letters, digits, and underscores (no spaces or symbols).
- Case-sensitive: `age` and `Age` are different variables.
- Cannot be a reserved keyword.
- Convention: use `snake_case` for variable names, e.g. `total_score`.

Checking and converting types

```
x = 10
print(type(x))      # <class 'int'>
y = str(x)          # convert int -> str: "10"
z = float(x)         # convert int -> float: 10.0
```

2. Keywords

Keywords are reserved words that form the syntax of Python itself. They have special meaning to the interpreter and cannot be used as variable, function, or class names.

Python keywords

False, None, True, and, as, assert, async, await, break, class, continue, def, del, elif, else, except, finally, for, from, global, if, import, in, is, lambda, nonlocal, not, or, pass, raise, return, try, while, with, yield.

Checking keywords in code

```
import keyword
print(keyword.kwlist)           # lists all keywords
print(keyword.iskeyword("for")) # True
print(keyword.iskeyword("data")) # False
```

Note: Built-in function names like `list`, `print`, or `str` are not keywords. They can technically be reassigned, but doing so is bad practice because it overrides the built-in.

3. Data Types

Every value in Python has a type. Python's built-in types fall into a few categories:

Category	Type(s)	Example
Numeric	int, float, complex	10, 3.14, 2+3j
Text	str	"hello"
Boolean	bool	True, False
Sequence	list, tuple, range	[1,2,3], (1,2,3), range(5)
Mapping	dict	{"a": 1}
Set	set, frozenset	{1, 2, 3}
None	NoneType	None

Mutable vs. immutable

- Immutable: int, float, bool, str, tuple, frozenset.
- Mutable: list, dict, set.

Type checking

```
print(type(10))           # <class 'int'>
print(type(3.14))         # <class 'float'>
print(type("hi"))         # <class 'str'>
print(type([1, 2]))        # <class 'list'>
print(type((1, 2)))        # <class 'tuple'>
print(type({"a": 1}))       # <class 'dict'>
print(type({1, 2}))        # <class 'set'>
print(type(None))          # <class 'NoneType'>
print(isinstance(10, int)) # True
```